Conclusions

This thesis work implemented and characterized an experimental apparatus for an Optical Dipole Trap (ODT) designed to trap a sample of ^{87}Rb atoms, which was integrated into a pre-existing setup based on a Magneto-Optical Trap (MOT) and optical molasses cooling. The ODT was realized using 1064 nm laser light coupled into a kagome Hollow-Core Photonic Crystal Fiber (HCPCF). A crucial objective of this work was the optimization and reliable light injection into the HCPCF. Standard injection procedures relying solely on power monitoring frequently resulted in light coupling into the silica jacket or into higher-order modes. To overcome this, a custom-built optical microscope was employed for near-field monitoring. On a test fiber, this approach enabled coupling into a fundamental LP_{01}-like mode with minor deformations, while also achieving $\sim 96\%$ transmission efficiency. The application of the same procedure to the HCPCF installed in the vacuum chamber yielded a final operating efficiency of $\sim 40\%$ with a partially asymmetric mode profile. This reduced performance on the in-chamber fiber is likely due to mechanical stresses applied to the fiber itself by the vacuum sealing system.

Once the ODT was operative, the experimental focus shifted to measurements inside the vacuum chamber. First, the molasses stage temperature was optimized through systematic investigation of the cooling beam detuning, the Acousto-Optic Modulator (AOM) ramp durations and amplitudes, and the cancellation of residual magnetic fields via compensation coils. These adjustments led to a minimum x-y-averaged atomic cloud temperature of $T = (9.6 \pm 0.1)$ μK.

The final characterization focused on the atom loading efficiency from the molasses into the ODT. The spatial overlap between the atomic cloud and the ODT beam was optimized by utilizing the compensation coils during the MOT phase to displace the atomic cloud from its optimal position, thereby increasing the overlap with the ODT. The successful isolation of the trapped atom signal relied on the implementation of a higher-resolution imaging system and, most importantly, on the mitigation of the intensity noise introduced by the AOM. This enhanced the signal-to-noise ratio, enabling the correct optimization of the ODT loading process, leading to a maximum observed capture efficiency of $(0.160 \pm 0.005)\%$.

Although numerically small, the observed capture efficiency is consistent with the order of magnitude predicted by the non-shallow trap theoretical model presented in this thesis. Applying the experimental parameters of the molasses and the trap geometry to the model yields an expected capture efficiency of $(0.41 \pm 0.04)\%$. The agreement on the order of magnitude suggests that the primary limitations to the capture efficiency are given by the intrinsic properties of the apparatus such as the distance between the fiber facet and the MOT center, the beam geometrical properties and the ODT optical power, together with the minimum temperature achieved. However, the experimental value remains approximately one-third of the theoretical expectation. It should be investigated whether this discrepancy comes from procedural artifacts such as the 600 μs delay between the end of the Time-of-Flight (TOF) sequence and the conclusion of the imaging flash.

Beyond addressing this discrepancy, the results obtained in this thesis establish a robust foundation for the future development of the experiment. The identification of the optimal parameters for both the optical molasses and the ODT loading sequence provides a reliable experimental baseline. This ensures that these operational stages are fully characterized and will not require fundamental recalibration in subsequent stages. Furthermore, the developed injection setup and near-field inspection protocol will be instrumental for introducing a second, counter-propagating 1064 nm beam. This configuration is necessary to realize an optical conveyor belt, enabling the efficient loading and deep confinement of atoms directly within the hollow core of the HCPCF.